Repetition Priming Effects in Proficient Mandarin–Cantonese and Cantonese–Mandarin Bidialectals: An Event-Related Potential Study

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Abstract

The present study adopted a repetition priming paradigm to investigate the bidialectal (bilingual) representation of speakers with different native dialects by event-related potential (ERP) technique. Proficient Mandarin–Cantonese and Cantonese–Mandarin bidialectals participated in the study. They were required to judge whether a word was a biological word or not, when the words (target word) were represented under four types of repetition priming conditions: Mandarin (prime)–Mandarin (target), Mandarin (prime)–Cantonese (target), Cantonese (prime)–Cantonese (target) and Cantonese (prime)–Mandarin (target). Results of reaction time and accuracy primarily indicated larger repetition priming effects in Mandarin–Mandarin and Cantonese–Cantonese (within-language) conditions than that in Mandarin–Cantonese and Cantonese–Mandarin (between-language) conditions. But more importantly, P200 and N400 mean amplitudes revealed distinct repetition priming effects between two types of participants. Specifically, both P200 and N400 indicated that the repetition priming effect in Mandarin–Mandarin condition was larger than that in Cantonese–Cantonese condition for Mandarin–Cantonese participants, whereas it was opposite for Cantonese–Mandarin participants. In addition, P200 also suggested opposite patterns of repetition priming effects in between-language priming conditions for two groups of participants. The repetition priming effect in Mandarin–Cantonese condition was larger than that in Cantonese–Mandarin condition for Mandarin–Cantonese participants, while for Cantonese–Mandarin participants, it was opposite (Mandarin–Cantonese < Cantonese–Mandarin). The results implied a clear asymmetric representation of two dialects for proficient bidialectals. They were further discussed in light of native dialect and language use frequency.

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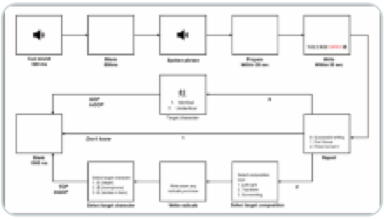

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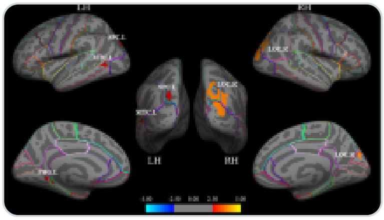

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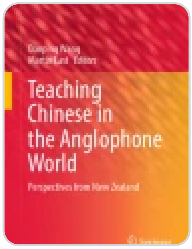

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Introduction

Bilinguals refer to individuals who can speak or use two languages. Traditional bilingual studies primarily focused on bilinguals with two languages from different systems, such as English and French or Chinese and English (e.g., Caramazza et al. 1973; Cutler et al. 1992; Barac and Bialystok 2012; Chee et al. 1999; Chen et al. 2015). But in addition to such type of bilinguals, there is another special type of bilinguals, that is bidialectals who can speak two dialects (Fabbro 1999). Dialect is a kind of variation form of a language (Ferguson 1959), such as Cantonese, a dialect in Chinese. Generally, different types of dialects in a language share a same writing system but differ in phonology systems. Thus as the special characteristic of dialects, an interesting question arose as to how bidialectal speakers organize their two types of dialects in brain.

In the study of bilinguals, several models have been proposed to illustrate the organization and the representation of two languages. They mainly discussed about the lexical and the conceptual representation of bilinguals' first and second language (L1 and L2). Such as the Word Association Model, it hypothesized that the lexical representation of L1 can access to the conceptual representation directly, while the lexical representation of L2 can only access to the conceptual representation via the lexical representation of L1 (Potter et al. 1984). On the contrary, the Concept Mediation Model assumed that the lexical representation of L1 and L2 access to the conceptual representation respectively, but there was no direct connection between L1 and L2 lexical representation (Potter et al. 1984). These two early models were considered to reflect the language presentation of less proficient and proficient bilinguals respectively (Potter et al. 1984; De Groot and Hoeks 1995; Mo et al. 2005; Li et al. 2006).

In the later model, the Revised Hierarchical Model (RHM), Kroll and Stewart (1990, 1994) further indicated that the connection from L1 lexical representation to L2 lexical representation was stronger than that from L2 lexical representation to L1 lexical representation. Moreover, the connection strength between L1 lexical representation and conceptual representation was stronger than that between L2 lexical representation and conceptual representation. In other words, the lexical and conceptual representations were asymmetrical for L1 and L2 (Sholl et al. 1995; Palmer et al. 2010; Chen et al. 2015). Moreover, based on the asymmetrical view, Heredia (1996) further proposed the Revision of the Revised Hierarchical Model. The model used more dominant language (MDL) and less dominant language (LDL) instead of L1 and L2 to illustrate the language representation. It emphasized that the use frequency of languages impacted bilinguals' language representation.

All the models mentioned above provided insight into bilingual representation. But do these models can explain bidialectal representation? Some researchers have tried to explore this issue by repetition priming paradigm. The repetition priming paradigm has been widely used in bilingual studies (e.g., Zeelenberg and Pecher 2003; Ulrich et al. 2013; Jouravlev et al. 2014). Especially with the development of neuroimaging methods, more and more studies combined this paradigm with event-related potential (ERP) and functional magnetic resonance imaging (fMRI) technique, to explore language representation and word recognition (e.g., Alvarez et al. 2003; Midgley et al. 2009; Geyer et al. 2011). The paradigm usually contained different types of priming conditions, such as within-language conditions (L1-L1, L2-L2) and between-language conditions (L1-L2, L2-L1). Researchers mainly focus on priming effects elicited by these priming conditions and investigated related issues. For example, Geyer et al. (2011)'s ERP study of proficient Russian-English bilinguals showed a symmetrical pattern of within-language and between-language repetition priming effects, which revealed the effect of L2 proficiency on the connectivity between lexical and conceptual representation of L1 and L2.

In the current studies of bidialectal representation, Ma et al. (2011)'s study of proficient Cantonese–Mandarin bidialectals found significant repetition priming effect in Mandarin–Cantonese priming condition, but they did not find such a priming effect in the opposite priming direction (Cantonese–Mandarin). The results showed an asymmetric between-language repetition priming effect between L1-L2 and L2-L1, which partially supported the Revised Hierarchical Model. Zhang and Zhang (2014) explored the bidialectal representation of proficient Cantonese–Mandarin bidialectals by auditory Cantonese and Mandarin words. For words with the same form but different meanings, the within-language priming effects of Mandarin and Cantonese were identical, but the between-language priming effects in Mandarin–Cantonese condition was larger than that in Cantonese–Mandarin condition. However, for words with different forms and meanings, the within-language priming effect of Cantonese was larger than that of Mandarin, whereas the priming effects in between-language priming conditions were similar. The results from different types of auditory words also supported the asymmetric view in the Revised Hierarchical Model. But in Quan (2012), the within-language repetition priming effects of Mandarin and Cantonese, and the between-language repetition priming effects in Mandarin–Cantonese and Cantonese–Mandarin priming conditions were both identical for proficient Cantonese–Mandarin bidialectals. Different from the former two studies, the study supported the Concept Mediation Model.

In addition to proficient bidialectals, Cai (2013)'s study of less proficient Mandarin–Cantonese bidialectals showed a repetition priming effect in Cantonese–Mandarin priming condition in semantic decision task, but did not show such effect in word decision task. They considered that the results supported the Word Association Model. But in Mai and Chen (2014), they found that less proficient Teochew-Cantonese bidialectals (Teochew is also a type of Chinese dialect) showed larger priming effects in Teochew-Cantonese priming condition than that in Cantonese–Teochew priming condition when prime and target words were translation equivalents. However, when prime and target words had related semantics, the between-language priming effects were similar. They thought that the results were consistent with the hypothesis of Revised Hierarchical Model.

As reviewed from the above studies, views on the language representation of speakers with two dialects were still controversial. One possible reason may be the unclear specific language background of bidialectals in previous studies, especially participants' native dialect and their use frequency of two dialects. For example, in studies of proficient speakers, researchers mainly explored bidialectals with Cantonese as native language (Ma et al. 2011; Zhang and Zhang 2014; Quan 2012), but did not consider speakers with Mandarin as native language. Moreover, as the language policy in mainland China, in which Mandarin is the official language and supposed to be used in many occasions, the use frequency of dialects may also affect the representation of bidialectal though speakers are proficient in both dialects.

Considering the potential problems mentioned above, we planned to further investigate the bidialectal representation with ERP technique in the present study, by both proficient Mandarin–Cantonese and Cantonese–Mandarin bidialectals. Conforming to most of previous studies, we adopted repetition priming paradigm with four types of repetition priming conditions, including within-language priming conditions, Mandarin (prime)–Mandarin (target) and Cantonese (prime)–Cantonese (target), and between-language priming conditions, Mandarin (prime)–Cantonese (target) and Cantonese (prime)–Mandarin (target). Since Cantonese and Mandarin shared a same writing system, we used language cues instead of masks, which have been widely used in priming studies (e.g., Dunabeitia et al. 2010; Midgley et al. 2009), before words. The language cues can signal the dialect types of the presented words and prevent potential translations or some other strategy used in completing the task. We would primarily observe the repetition priming effects reflected by reaction time, accuracy and two ERP components: P200 and N400. Specially, P200 is distributed in frontal-central area and reaches to the peak at around 200ms after stimuli onset. It is considered to be related with early recognition of stimuli (Cheng et al. 2014). N400 is thought to be related to semantic processing. It is usually detected in parietal areas and peaks in 250–550ms time window after stimulus is represented (e.g., Kounios and Holcomb 1992; Kutas and Federmeier 2011). These two components are usually used in priming studies (e.g., Misra and Holcomb 2003; Morris et al. 2007; Shuai and Gong 2014).

We hypothesized that all four types of priming conditions would elicit priming effects in reaction time, accuracy, P200 and N400. Moreover, if repetition priming effects were different between priming conditions, we would observe different reaction time, accuracy, P200 or N400 mean amplitudes between these conditions; whereas if they were similar, no difference would be observed. Furthermore, if the native dialect of bidialectals affected repetition priming effects in certain priming condition, we would find different reaction time, accuracy, P200 or N400 mean amplitudes between Mandarin–Cantonese and Cantonese–Mandarin participants; while it had no influence on repetition priming effects, no distinction would be found.

Method

Participants

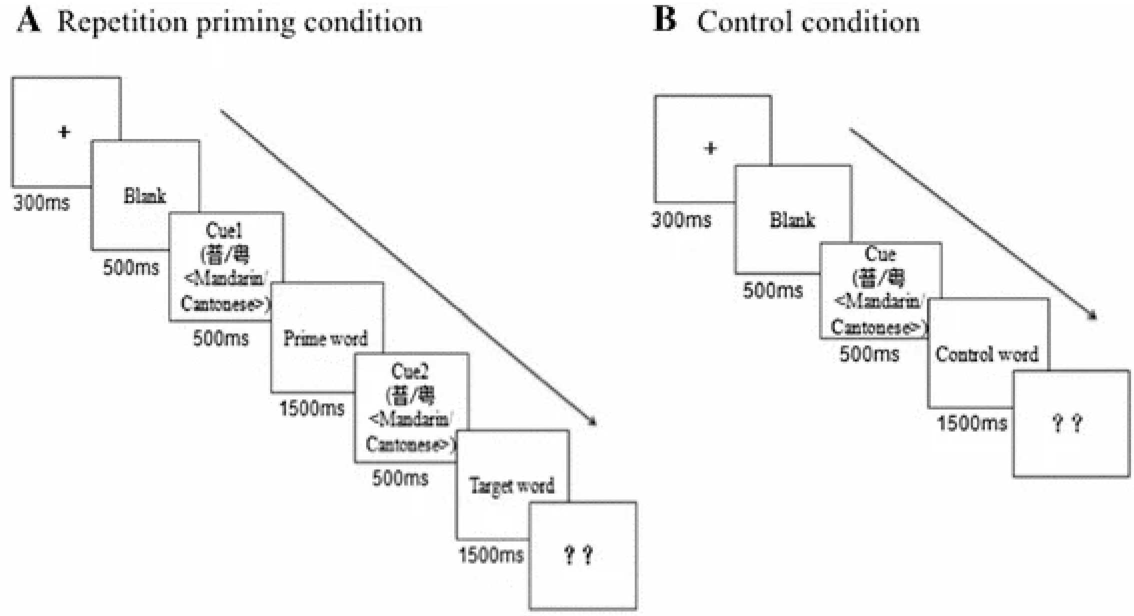
Thirty undergraduate students took part in the experiment, including 15 Mandarin–Cantonese bidialectals from mainland China studied in Hong Kong or Macou (L1: Mandarin, L2: Cantonese) (9 female, age ranging from 22 to 25, mean = 23.1, SD = 1.2), and 15 Cantonese–Mandarin bidialectals from Hong Kong or Macou studied in mainland China (L1: Cantonese, L2: Mandarin) (8 female, ageranging from 22 to 26, mean = 24.3, SD = 1.7). The two groups of participants (Mandarin–Cantonese and Cantonese–Mandarin bidialectals) have no significant difference in Mandarin and Cantonese listening, speaking, reading and writing according to their self-evaluations based on a seven-point self-rating scale (1 = unable; 7 = expert) (as shown in Table 1). However, their use frequency of Mandarin and Cantonese were significantly different. The self-evaluation of Cantonese use frequency for Mandarin–Cantonese bidialectals was higher than that of Mandarin use frequency (Mandarin: 3.33, Cantonese: 5.93, t = −9.342, $p < 0.05$), whereas for Cantonese–Mandarin bidialectals, the self-evaluation of Cantonese use frequency was lower than that of Mandarin use frequency (Mandarin: 6.20, Cantonese: 3.27, t = -11.641, $p < 0.05$). All the participants were right-handed according to their self-evaluations and had normal or corrected-to-normal visual acuity. They had no history of neurological or language disability. The experiment was approved by the ethics committee of Jinan University First Affiliated Hospital in Guangzhou, China. The participants gave informed consent before taking part in the experiment and received payments for their participation.

Table 1 Language background of Mandarin–Cantonese and Cantonese–Mandarin bidialectals

Materials

The experimental materials consisted of 120 Mandarin disyllabic words, 120 Cantonese disyllabic words and 120 control words. Specifically, both Mandarin and Cantonese disyllabic words included 60 biological words (e.g., Mandarin: 黄牛(cattle); Cantonese: BB虾(infant)), and 60 non-biological words (e.g., Mandarin: 爆竹(firecracke); Cantonese: 藤条(rattan)). The control words included 60 Mandarin disyllabic words (e.g., 老师(teacher)) and 60 Cantonese disyllabic words (e.g., 晒士(size)). The Mandarin words were selected from *Contemporary Chinese Dictionary*, and the Cantonese words were selected from *Cantonese Dialect Dictionary*. They were both written in simplified characters, and were matched in stroke number (Mandarin: 20.8, Cantonese: 23.9, t = −2.084, $p > 0.05$) and use frequency (Mandarin: 0.937, Cantonese: 0.909, t= $2.954, p > 0.05$). In addition, the familiarity of Mandarin and Cantonese words were evaluated by 10 proficient bidialectal speakers of Mandarin and Cantonese (came from Jinan University undergraduates and did not participated in the ERP experiment), based on a five-point self-rated scale (1 = unfamiliar, 5 = very familiar). The results showed no significant difference between the familiarity of Mandarin and Cantonese words (Mandarin: 4.7, Cantonese: 4.4, t = 1.342, $p > 0.05$).

Fig. 1



Experimental procedures for repetition priming condition (**a**) and control condition (**b**)

Procedure

We adopted an unmasked repetition priming paradigm with four types of repetition priming conditions in the experiment, including (1) Mandarin (prime)–Mandarin (target) (MM, e.g.,<蜜蜂 honeybee>蜜蜂 -<honeybee>)(2) Cantonese (prime) - Cantonese (target) (CC, e.g.,雪条 <ice cream>雪条 -<ice cream>); (3) Mandarin (prime) -Cantonese (target) (MM, e.g.,父亲 <father>-老豆 <father>) and (4) Cantonese (prime) - Mandarin (target) (CM, e.g.,纸鹞 <kite>-风筝 <kite>) (see more details in Appendix). Each condition contained 30 biological word pairs and 30 non-biological word pairs. In addition to priming conditions, there were two types of control conditions in the experiment. They were Mandarin control condition (M, e.g., 毛巾 <towel>) and Cantonese control condition (C, e.g., 古仔 <story>) respectively, and each has 60 words.

The procedures for a repetition priming and a control trial were as shown in Fig. 1a, b. Compared with repetition priming trial, the control trial did not contain a prime word. As both Mandarin and Cantonese words were written in simplified characters, there would be a language cue prior to the prime/target/control word for the participants. Participants were instructed to read the word according to the cue silently (the purpose of silent reading was to minimize the effect of potential translation or some other strategy in completing the experimental task), and when they saw "??" in the screen, they should judge whether the presented word was a biological word or not. For half of the participants, if the presented word was a biological word, they should press "f" in the keyboard while if it was not, they should press "j", and for the other half, the responses were opposite. The repetition priming and control conditions would be presented randomly to the participants.

The stimuli were presented by E-Prime 2.0 on an IBM computer. Participants would practice several trials to familiarize the procedure and then complete the formal experiment. The formal experiment consisted of four blocks, and each block contained 90 trials. Participants would take a rest for two minutes between blocks. The whole experiment lasted approximately 40 min.

Electroencephalogram (EEG) Recording

The EEG was recorded by a 64-channel (Ag–AgCl) Brain Products (BP). A BrainAmp amplifier was used by BrainVision Recorder software (version 1.20). Electrodes were based on the standard international 10–20 system convention. Electrode Fpz was served as the ground and electrode FCz was recorded for reference. The vertical electrooculogram (EOG) was recorded infra-orbitally from the left eye. The horizontal EOG was recorded in right orbital rim. The impedance of each electrode was kept below $5\ \mathrm{k}\Omega$. EEG and EOG signals were digitized online at 500 Hz and band-pass filtered from 0.01 to 15 Hz.

Data Analysis

Behavioral Data

We first calculated the reaction time (RT) and accuracy (ACC) of repetition priming conditions and control conditions, and then compared the RT and ACC between repetition priming conditions and their corresponding control conditions (based on the dialect type of target word, e.g., MM vs. M) by t-tests to confirm that the repetition priming conditions did elicit priming effects.

If the priming effects did exist in RT and/or ACC, we would further calculate the repetition priming effects by subtracting the RT and/or ACC of repetition priming conditions from that of their corresponding control conditions. For example, the repetition priming effect in RT of Mandarin–Mandarin condition would be obtained by subtracting the RT of Mandarin–Mandarin condition from Mandarin control condition. At last, the repetition priming effects of RT and/or ACC would be analyzed by 2 (native language: Cantonese/Mandarin) × 4 (priming condition: MM/MC/CC/CM) mixed repeated ANOVA with native language as a between-subject factor and repetition type as a within-subject factor. Participants (F_1) and items (F_2) were both considered as random factors in the analyses.

ERP Data

EEG data would be pre-processed by BrainVision Analyzer software (version 2.0) offline firstly. At first, electrode FCz was converted to bilateral mastoids (A1, A2) as new references, and then the eye blinks were corrected and the artifacts exceeding $\pm 80\ \mathbf{mv}$ were rejected. After that, epochs were segmented ranging from 100 ms before the target word onset and 600 ms after the target word onset. Then the baseline correction was conducted to correct the pre-stimulus activities. At last, we got the grand average waveforms of four types of repetition priming conditions and two types of control conditions for the two groups of participants.

According to the grand average waveforms, we selected 150–250 ms after the stimuli onset as the time window for P200 and 250–400 ms after the stimuli onset as the time window for N400. The mean amplitudes of P200 and N400 in repetition priming and control conditions were calculated by averaging the amplitudes of electrode F3, F4, FZ, C3, C4, CZ, P3, P4 and PZ in their corresponding time windows. Then as in analyzing behavioral data, we first conducted t-tests to confirm whether the repetition priming effects of P200 and/or N400 did exist or not. Then we calculated the repetition priming effects of P200 and/or N400 by subtracting repetition priming conditions from their corresponding control conditions respectively. Lastly, the repetition priming effects of P200 and/or N400 would be conducted by two-way mixed measure ANOVA with native language (Cantonese/Mandarin) as a between-subject factor and priming condition (MM/MC/CC/CM) as a within-subject factor. Both participants (F_1) and items (F_2) were considered as random factors in the analyses.

Table 2 Reaction time (RT)and accuracy (ACC) of different kinds of repetition types for two groups (ms) $(\mathrm{M} \pm \mathrm{SD})$

Results

One participant from Mandarin–Cantonese group and two participants from Cantonese–Mandarin group were eliminated in the statistical analyses for their accuracy being lower than 80%. The behavioral and ERP results of the remained 14 Mandarin–Cantonese and 13 Cantonese–Mandarin participants were as follows:

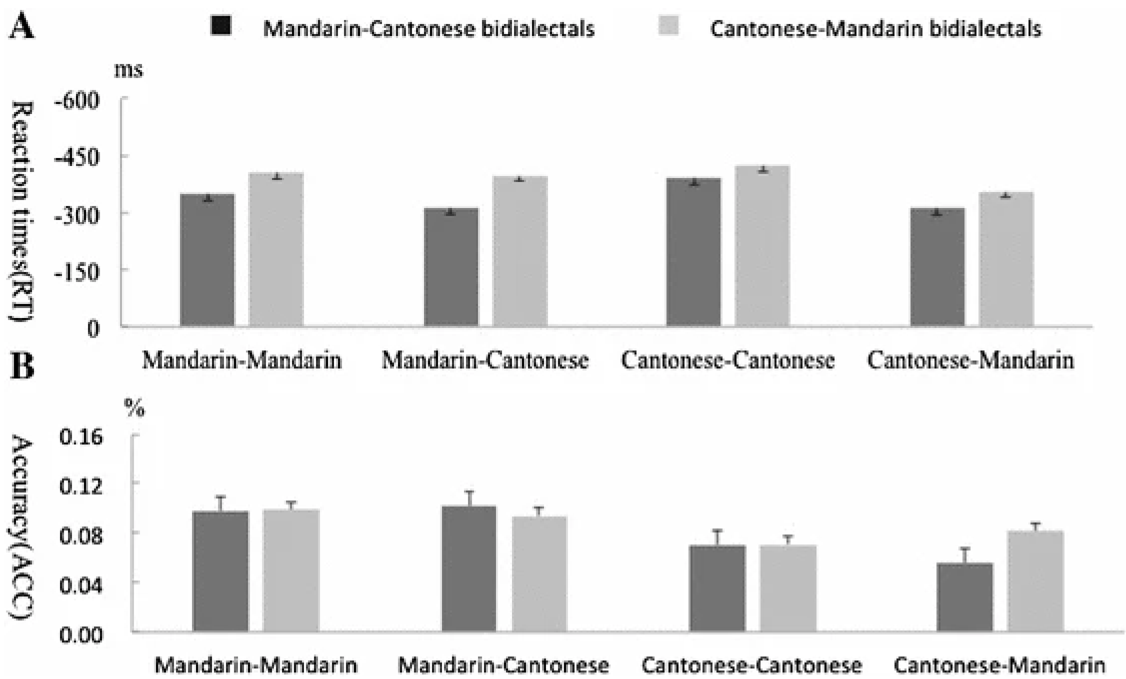
Behavioral Results

RT

Table 2 showed the results of RT in four types of repetition priming conditions and Mandarin and Cantonese control conditions. The t-tests of RT between repetition priming conditions and control conditions showed significant differencebetween the repetition priming conditions and their corresponding control conditions respectively (Mandarin–Cantonese bidialectals: MM vs. M : t = -8.220, $p < 0.05$; MC vs. C: t = -11.782, $p < 0.05$; CC vs. C: t = -10.598, $p < 0.05$; CM vs. M : t = -6.586, $p < 0.05$. Cantonese–Mandarin bidialectals: MM vs. M: t = -7.191, $p < 0.05$; MC vs. C: t = -11.899, $p < 0.05$; CC vs. C: t = -12.741, $p < 0.05$; CM vs. M : t = -7.203, $p < 0.05$). In addition, no difference between control conditions for both groups of participants was found (Mandarin–Cantonese bidialectals: M vs. C: t = -0.433, $p > 0.05$; Cantonese–Mandarin bidialectals: M vs. C: t = -0.088, $p > 0.05$). The results confirmed that the four types of repetition priming conditions did elicit repetition priming effects of RT.

The further two-way mixed measure ANOVA of repetition priming effects in RT (Fig. 2a) showed that the main effect of priming condition was significant by participants ($F_1(1,3) = 6.44, p < 0.05, \eta_1^2 = 0.205; F_2(1,3) = 1.598, p > 0.05, \eta_2^2 = 0.091$). The post hoc test showed that the priming effects in MM was significantly larger than that in CM condition, and the priming effects in CC condition was significantly larger than that in MC condition and CM condition $(ps < 0.05)$. There were no significant differences between the priming effects of other priming conditions $(ps > 0.05)$. The main effect of native language was only significant by items $(F_1(1,1) = 2.145, p > 0.05, \eta_1^2 = 0.079; F_2(1,1) = 8.528, p < 0.05, \eta_2^2 = 0.151)$, and the interaction between native language and priming condition was not significant ($F_1(1,3) = 0.832, p > 0.05, \eta_1^2 = 0.032; F_2(1,3) = 0.395, p > 0.05, \eta_2^2 = 0.024$).

Fig. 2



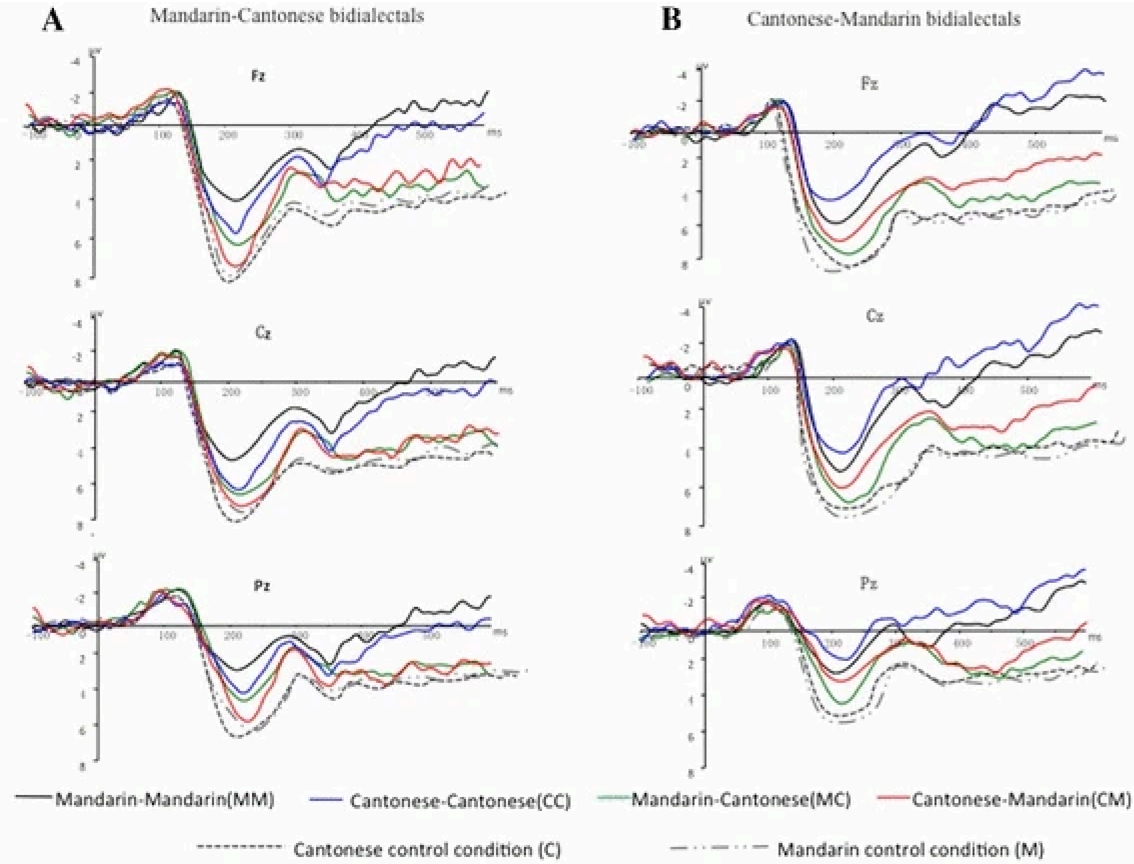
Repetition priming effects of reaction time (RT) (**a**)and original accuracy(ACC) (**b**) of four types of repetition priming conditions for Mandarin–Cantonese and Cantonese–Mandarin bidialectals (the line segment represents one standard error)

ACC

The t-tests of ACC also showed significant difference between repetition priming conditions and their corresponding control conditions (Mandarin–Cantonese bidialectals: MM vs. M: t = 5.297, $p < 0.05$; MC vs. C: t = 6.559, $p < 0.05$; CC vs. C: t = 6.069, $p < 0.05$; CM vs. M: t = 2.664, $p < 0.05$. Cantonese–Mandarin bidialectals: MM vs. M: t = 4.330, $p < 0.05$; MC vs. C: t = 6.673, $p < 0.05$; CC vs. C: t = 4.721, $p < 0.05$; CM vs. M: t = 3.418, $p < 0.05$). But there were no difference between control conditions for both groups of participants (Mandarin–Cantonese bidialectals: M vs. C: t = 1.925, $p > 0.05$; Cantonese–Mandarin bidialectals: M vs. C: t = -0.726, $p > 0.05$). The results suggested that the repetition priming conditions also elicited priming effects in ACC.

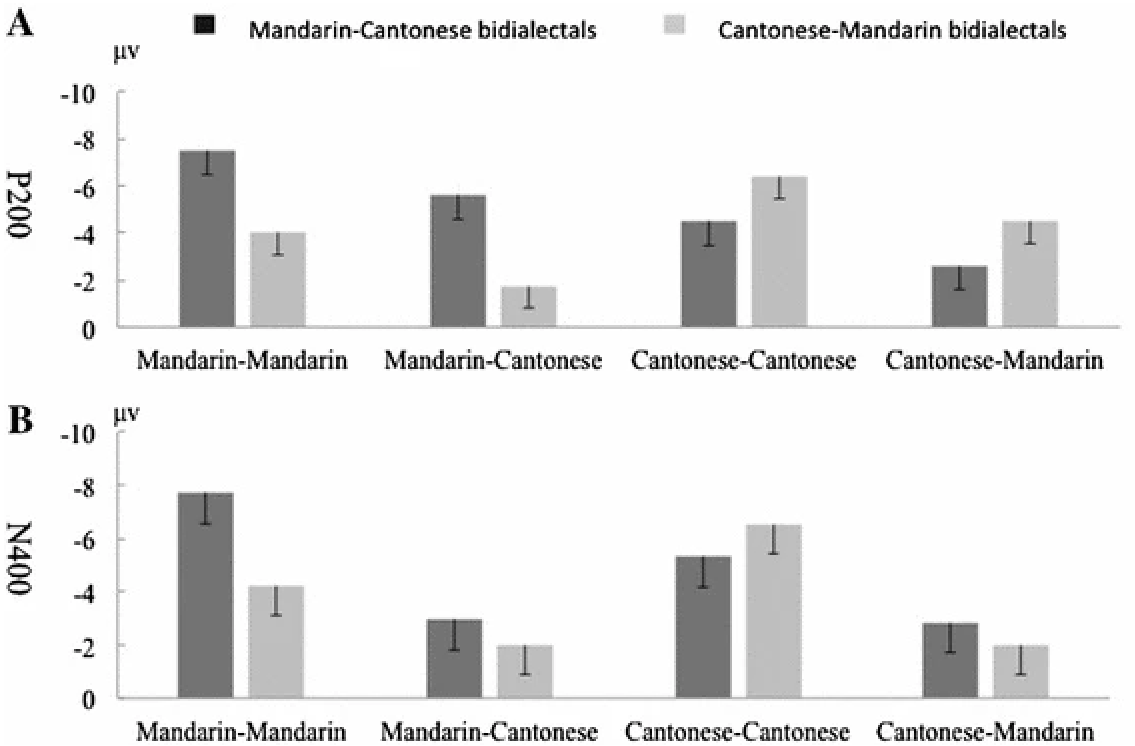
The further ANOVA of priming effects in ACC (Fig. 2b) showed that the main effect of priming condition was significant ($F_1(1,3) = 6.296, P < 0.05, \eta_1^2 = 0.201$; $F_2(1,3) = 2.861, P < 0.05, \eta_2^2 = 0.152$). The results of post hoc test showed that the priming effects in MM condition was significantly larger than that in CC and CM condition ($ps < 0.05$), the priming effects in MC condition was significantly larger than that in CC condition and CM condition ($ps < 0.05$). There were no significant differences between the priming effects of other priming conditions $(ps > 0.05)$. The main effect of native language was not significant ($F_1(1,1) = 0.105, p > 0.05, \eta_1^2 = 0.004$; $F_2(1,1) = 0.251, p > 0.05, \eta_2^2 = 0.005$). The interaction between native language and priming condition was not significant ($F_1(1,3) = 1.266, p > 0.05, \eta_1^2 = 0.048$; $F_2(1,3) = 0.629, p > 0.05, \eta_2^2 = 0.038$).

Fig. 3



Grandaverage waveforms of repetition priming conditions (Mandarin–Mandarin (MM); Cantonese–Cantonese (CC); Mandarin–Cantonese (MC), Cantonese–Mandarin (CM)) and control conditions (Mandarin (M); Cantonese (C)) for Mandarin–Cantonese bidialectals (**a**) and Cantonese–Mandarin bidialectals (**b**)

Fig. 4



Repetition priming effects of P200 (**a**) and N400 (**b**) in repetition priming conditions (Mandarin–Mandarin (MM); Cantonese–Cantonese (CC); Mandarin–Cantonese (MC), Cantonese–Mandarin (CM)) for Mandarin–Cantonese and Cantonese–Mandarin bidialectals (the line segment represents one standard error)

ERP Results

The grand average waveforms of repetition priming conditions and control conditions were shown in Fig. 3 (A: Mandarin–Cantonese bidialectals; B: Cantonese–Mandarin bidialectals). As shown in the figures, we can find clear P200 and N400 for both groups of participants. The followings were the statistical results of the mean amplitudes of P200 and N400.

P200 Mean Amplitude

Firstly, the t-tests of P200 between repetition priming conditions and control conditions suggested that the four types of repetition priming conditions did elicit repetition priming effects of P200 (Mandarin–Cantonese bidialectals: M vs. C: t = 0.193, $p > 0.05$; MM vs. M : t = −11.344, $p < 0.05$; MC vs. C: t = −4.412, $p < 0.05$; CC vs. C: t = −6.208, $p < 0.05$; CM vs. M : t = -2.377, $p < 0.05$. Cantonese–Mandarin bidialectals: M vs. C: t = −0.311, $p > 0.05$; MM vs. M: t = −5.568,$p < 0.05$; MC vs. C: t = −3.169, $p < 0.05$; CC vs. C: t = −9.257, $p < 0.05$; CM vs. M: t = −3.739, $p < 0.05$).

The further ANOVA results of P200 repetition priming effects (Fig. 4a) revealed a significant main effect of priming condition ($F_1(1,3) = 12.647$, $P < 0.05$, $\eta_1^2 = 0.336$; $F_2(1,3) = 5.700$, $P < 0.05$, $\eta_2^2 = 0.263$). The results of post hoc test showed that the repetition priming effects in MM was significantly larger than that in MC and CM, and the repetition priming effects in CC was significantly larger than that in MC and CM ($ps < 0.01$). There were no significant differences between the repetition priming effects of other priming conditions ($ps > 0.05$). The main effect of native language was only significant by items ($F_1(1,1) = 1.722$, $P > 0.05$, $\eta_1^2 = 0.064$; $F_2(1,1) = 4.258$, $P < 0.05$, $\eta_2^2 = 0.081$).

The interaction between native language and priming condition was significant ($F_1(1,3) = 24.894, p < 0.05, \eta_1^2 = 0.499$; $F_2(1,3) = 13.961, p < 0.05$, $\eta_2^2 = 0.466$). Simple effect test revealed that for native Mandarin, the repetition priming effects in MM was significantly larger than that in CC and CM, and MC was significantly larger than CM ($ps < 0.05$). No significant difference was found between other priming conditions ($ps > 0.05$). However, for native Cantonese, the repetition priming effects in MM was significantly smaller than that in CC, and MC was significantly smaller than CM and CC ($ps < 0.05$). There were no significant differences between other priming conditions ($ps > 0.05$).

N400 Mean Amplitude

The t-tests results of N400 suggested that the four types of repetition priming conditions did elicit repetition priming effects of N400 (Mandarin–Cantonese bidialectals: M vs. C: t = −0.144, $p > 0.05$; MM vs. M: t = −11.587, $p < 0.05$; MC vs. C: t = −2.681, $p < 0.05$; CC vs. C: t = −6.706, $p < 0.05$; CM vs. M: t = −4.730, $p < 0.05$. Cantonese–Mandarin bidialectals: M vs. C: t = 0.143, $p > 0.05$; MM vs. M: t = −6.489, $p < 0.05$; MC vs. C: t = −2.851, $p < 0.05$; CC vs. C: t = −7.410, $p < 0.05$; CM vs. M: t = −3.385, $p < 0.05$).

The further ANOVA results of N400 repetition priming effects (Fig. 4b) showed a significant main effect of priming condition ($F_1(1,3) = 43.073, p < 0.05$, $\eta_1^2 = 0.633$; $F_2(1,3) = 18.215, p < 0.05$, $\eta_2^2 = 0.532$). The post hoc test showed that the repetition priming effects of N400 in MM was significantly larger than that in MC and CM, and the repetition priming effects of N400 in CC was significantly larger than that in MC and CM ($ps < 0.05$). There were no significant differences between the repetition priming effects of other priming conditions ($ps > 0.05$). The main effect of native language was not significant ($F_1(1,1) = 4.093, p > 0.05, \eta_1^2 = 0.141$; $F_2(1,1) = 2.111, p > 0.05, \eta_2^2 = 0.042$).

The interaction between native language and priming condition was significant ($F_1(1,3) = 9.844, p < 0.05, \eta_1^2 = 0.283; F_2(1,3) = 2.278, p > 0.05, \eta_2^2 = 0.125$). Simple effect test revealed that for native Mandarin, the N400 repetition priming effects in MM was significantly larger than that in three other priming conditions ($ps < 0.05$), and that in CC was significantly larger than that in CM and MC ($ps < 0.05$). No significant difference was found between the repetition priming effects of other priming conditions ($ps > 0.05$). While for native Cantonese, the repetition priming effects in MM was significantly larger than that in MC and CM conditions($ps < 0.05$), while the repetition priming effects in CC was significantly larger than that in MC, CM and MM ($ps < 0.05$). There were no significant differences between the repetition priming effects in other priming conditions ($ps > 0.05$).

Discussion

The present repetition priming study of proficient Mandarin–Cantonese and Cantonese–Mandarin bidialectals showed repetition priming effects of RT, ACC, P200 and N400 mean amplitudes in all four types of priming conditions. Specifically, the priming effect of RT in within-language repetition priming conditions (MM and CC) were significantly larger than that in between-language repetition priming conditions (MC and CM). The similar results of RT were also observed in many previous studies (e.g. Dufour and Kroll 1995; Cheung and Chen 1998; Phillips et al. 2006), which reflected the faster lexical access of target words in within-language priming conditions. However, we found that no matter the target words were Mandarin or Cantonese words, the priming effects of accuracy were always higher in the conditions when Mandarin words were the priming words. The reason for it may be that we used simplified characters to represent both Mandarin and Cantonese words in the experiment. But in daily life, Cantonese words are also presented with traditional or some other special characters. Thus though both groups of participants reported that they were equally proficient in both dialects' reading, they may process the form of Mandarin words more efficiently.

As seen from the behavioral results, both RT and ACC showed similar differences of priming effects between priming conditions for two groups of participants. However, the priming effects of P200 and N400 mean amplitude not only showed different priming effects between within-language and across-language conditions, but also revealed distinctions between Mandarin–Cantonese and Cantonese–Mandarin bidialectals.

Firstly in P200, Mandarin–Cantonese participants showed a larger priming effect in MM condition than that in CC condition. In addition, the effect in MC condition was larger than that in CM condition. On contrast, Cantonese–Mandarin participants presented an opposite pattern, that is, the priming effect in MM condition was smaller than that in CC condition, and the priming effect in MC condition was smaller than that in CM condition. The results of P200 clearly revealed the influence of native language on priming effects. In addition, P200 has been considered to reflect the early processing of lexical forms (e.g. Dom Nguez et al. 2006; Guo et al. 2012). Thus we can further infer from the results of between-language priming conditions, that the connection from L1 lexical representation to L2 lexical representation is stronger than the reverse connection.

With regard to N400, we also found similar difference, as in P200, between within-language priming conditions for proficient Mandarin–Cantonese and Cantonese–Mandarin bidialectals. But we did not find significant difference in between-language priming effects for both of them. As N400 is primarily related to semantic processing (e.g. Lanwermeyer et al. 2016; Midgley et al. 2009), the similar effects in L1-L2 and L2-L1 may because of the stronger connection between L1 lexical representation and conceptual representation. This stronger connection could compensate the smaller priming effect of L2 lexical form on the processing of L1 lexical form, which we predicted in former P200 results, and facilitated the lexical access of L1 words in L2-L1.

As discussed above, the present study clearly implied an asymmetric pattern for the representation of bidialectal. This is consistent with the studies of proficient bidialectals we mentioned above (Ma et al. 2011; Zhang and Zhang 2014), and many studies on traditional bilinguals (e.g., Dufour and Kroll 1995; Palmer et al. 2010; Chen et al. 2015). But more importantly, we further indicated the effect of native dialect on the representation of first and second dialect.

Furthermore, we considered the use frequency of two groups of participants. According to their self-report, though they were proficient in both dialects to approximately the same extend, they used two dialects with different frequencies. Mandarin–Cantonese participants used Cantonese more frequently in their daily life while Cantonese–Mandarin participants used Mandarin more frequently. Thus we can consider that Cantonese may be a more dominant language for Mandarin–Cantonese participants but for Cantonese–Mandarin participants, Mandarin may become more dominant. However, as has indicated in the results, the use superiority of L2 did not seem to alter the representation of first and second dialect. It indicated that native dialect may have a stronger influence than use frequency on the representation of two dialects for proficient bidialectals. In addition, the results also seemed inconsistent with Heredia's revised version of Revised Hierarchical Model based on more dominant and less dominant languages.

Conclusion

The study investigated the bidialectal representation of proficient Mandarin–Cantonese and Cantonese–Mandarin bidialectals by repetition priming effects. The results clearly showed an asymmetric pattern for the representation of first and second dialect, which was consistent with the asymmetric hypothesis in Revised Hierarchical Model. Moreover, the results also revealed a significant role of native language (dialect) on bidialectal representation, though the use frequency of native language (dialect) was lower than that of second language (dialect).

References

Alvarez, R. P., Holcomb, P. J., & Grainger, J. (2003). Accessing word meaning in two languages: An event-related brain potential study of beginning bilinguals. *Brain & Language*, *87*(2), 290–304.

Barac, R., & Bialystok, E. (2012). Bilingual effects on cognitive and linguistic development: Role of language, cultural background, and education. *Child Development*, *83*(2), 413–422. doi:10.1111/j.1467-8624.

Cai, L. J. (2013). The Long-term cross-language repetition priming of less proficient Cantonese–Mandarin bilinguals: An ERP Study (Doctoral dissertation, Jinan University).

(In Chinese)

Caramazza, A., Yeni-Komshian, G. H., Zurif, E. B., & Carbone, E. (1973). The acquisition of a new phonological contrast: The case of stop consonants in French–English bilinguals. *The Journal of the Acoustical Society of America*, *54*(2), 421–428.

Chee, M. W., Tan, E. W., & Thiel, T. (1999). Mandarin and English single word processing studied with functional magnetic resonance imaging. *Journal of Neuroscience*, *19*(8), 3050–3056.

Cheng, X., Schafer, G., & Riddell, P. M. (2014). Immediate auditory repetition of words and nonwords: An ERP study of lexical and sublexical processing. *PLoS One*, *9*(3), e91988.

Chen, C. Y., Zhang, X. X., Li, L., & Wang, R. M. (2015). Bilingual memory representations in less fluent Chinese–English bilinguals: An event-related potential study. *Psychological Reports*, *116*(1), 230–241. doi:10.2466/28.PR0.116k13w7.

Cheung, H., & Chen, H. C. (1998). Lexical and conceptual processing in Chinese–English bilinguals: Further evidence for asymmetry. *Memory & Cognition*, *26*(5), 1002–1013.

Cutler, A., Mehler, J., Norris, D., & Segui, J. (1992). The monolingual nature of speech segmentation by bilinguals. *Cognitive Psychology*, *24*(3), 381–410.

De Groot, A. M. B., & Hoeks, J. C. (1995). The development of bilingual memory: Evidence from word translation by trilinguals. *Language Learning*, *45*(4), 683–724.

Dom Nguez, A., Alija, M., Cuetos, F., et al. (2006). Event related potentials reveal differences between morphological (prefixes) and phonological (syllables) processing of words. *Neuroscience Letters*, *408*(1), 10–15. doi:10.1016/j.neulet.2006.06.048.

Dufour, R., & Kroll, J. F. (1995). Matching words to concepts in two languages: A test of the concept mediation model of bilingual representation. *Memory & Cognition*, *23*(2), 166–180.

Dunabeitia, J. A., Perea, M., & Carreiras, M. (2010). Masked translation priming effects with highly proficientsimultaneous bilinguals. *Experimental Psychology*, *57*, 98–107.

Fabbro, F. (1999). Handbook of neurolinguistics. *Neuropsychologia*, *37*(1), 119–121.

Ferguson, C. A. (1959). Diglossia. *Word*, *15*, 325–340.

Geyer, A., Holcomb, P. J., Midgley, K. J., & Grainger, J. (2011). Processing words in two languages: An event-related brain potential study of proficient bilinguals. *Journal of Neurolinguistics*, *24*(3), 338–351.

Guo, T., Misra, M., & Tam, J. W. (2012). On the time course of accessing meaning in a second language: An electrophysiological and behavioral investigation of translation recognition. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *38*(5), 1165–1186. doi:10.1037/a0028076.

Heredia, R. R. (1996). Bilingual memory: A re-revised version of the hierarchical model of bilingual memory. *CRL Newsletter*, *10*(3), 3–10.

Hernandez, A. E., Bates, E. A., & Avila, L. X. (1996). Processing across the language boundary: A cross-modal priming study of Spanish–English bilinguals. *Journal of Experimental Psychology Learning Memory & Cognition*, *22*(4), 846–864.

Jouravlev, O., Lupker, S. J., & Jared, D. (2014). Cross-language phonological activation: Evidence from masked onset priming and ERPs. *Brain and Language*, *134*, 11–22.

Kounios, J., & Holcomb, P. J. (1992). Structure and process in semantic memory: Evidence from event-related brain potentials and reaction times. *Journal of Experimental Psychology General*, *121*(4), 459–479.

Kroll, J. F., & Stewart, E. (1990). Concept mediation in bilingual translation. *Bulletin of the Psychonomic Society*, *28*(6), 510.

Kroll, J. F., & Stewart, E. (1994). Category interference in translation and picture naming: Evidence for asymmetric connections between bilingual memory representations. *Journal of memory and language*, *33*(2), 149–174.

Kutas, M., & Federmeier, K. D. (2011). Thirty years and counting: Finding meaning in the N400 component of the event-related brain potential (ERP). *Annual Review of Psychology*, *62*, 621–647. doi:10.1146/annurev.psych.093008.131123.

Lanwermeyer, M., Henrich, K., Rocholl, M. J., et al. (2016). Dialect variation influences the phonological and lexical-semantic word processing in sentences electrophysiological evidence from a cross-dialectal comprehension study. *Frontiers in Psychology*, 7, 739. doi:10.3389/fpsyg.2016.00739.

Li, L., Mo, L., Wang, R. M., & Luo, X. Y. (2006). The effect of cross-language repetition priming in less proficient Chinese–English bilinguals. *Psychological Science*, *38*(5), 672–680. (In Chinese).

Mai, S. Y., & Chen, J. (2014). The semantic access of less-proficient Teochew–Cantonese bilinguals: Evidences from processing of spoken words. *Psychological Science*, *46*(2), 227–237. (In Chinese).

Ma, L. J., Wei, W., & Zhang, J. J. (2011). The effect of repetition priming in highly proficient Mandarin–Cantonese bilinguals. *Psychological Research*, 4(1), 24–29. (In Chinese).

Midgley, Katherine J., Holcomb, Phillip J., & Grainger, Jonathan. (2009). Masked repetition and translation priming in second language learners: A window on the time-course of form and meaning activation using ERPs. *Psychophysiology*, *46*(3), 551–565. doi:10.1111/j.1469-8986.2009.00784.x.

Midgley, K. J., Holcomb, P. J., & Jonathan, G. (2009). Masked repetition and translation priming in second language learners: A window on the time-course of form and meaning activation using ERPs. *Psychophysiology*, *46*(3), 551–565. doi:10.1111/j.1469-8986.2009.00784.x.

Misra, M., & Holcomb, P. J. (2003). Event-related potential indices of masked repetition priming. *Psychophysiology*, *40*(1), 115–130.

Mo, L., Li, L., & Wang, R. M. (2005). Evidence for long-term cross-language repetition priming of the highly proficient Chinese–English bilinguals. *Psychological Science*, *28*(6), 1288–1293. (In Chinese).

Morris, J., Franck, T., Grainger, J., & Holcomb, P. J. (2007). Semantic transparency and masked morphological priming: An ERP investigation. *Psychophysiology*, 44, 506–521. doi:10.1111/j.1469-8986.2007.00538.x.

Palmer, S. D., van Hooff, J. C., & Havelka, J. (2010). Language representation and processing in fluent bilinguals: Electrophysiological evidence for asymmetric mapping in bilingual memory. *Neuropsychologia*, *48*(5), 1426–1437. doi:10.1016/j.neuropsychologia.2010.01.010.

Phillips, N. A1., Klein, D., Mercier, J., & de Boysson, C. (2006). ERP measures of auditory word repetition and translation priming in bilinguals. *Brain Research*, *1125*(1), 116–131. doi:10.1016/j.brainres.2006.10.002.

Potter, M. C., So, K. F., Eckardt, B. V., & Feldman, L. B. (1984). Lexical and conceptual representation in beginning and proficient bilinguals. *Journal of Verbal Learning and Verbal Behavior*, *23*(1), 23–38.

Quan, J. J. (2012). The Repetition priming of proficiency Cantonese–Mandarin bilinguals: An ERP study (Doctoral dissertation, Jinan University). **(In Chinese)**

Sholl, A., Sankaranarayanan, A., & Kroll, J. F. (1995). Transfer between picture naming and translation: A test of asymmetries in bilingual memory. *Psychological Science*, *6*(1), 45–49.

Shuai, L., & Gong, T. (2014). Temporal relation between top-down and bottom-up processing in lexical tone perception. *Frontiers in Behavioral Neuroscience*, *79*(8), 97–97. doi:10.1016/j.brainres.2006.10.002.

Ulrich, M., Hoenig, K., Gron, G., & Kiefer, M. (2013). Brain activation during masked and unmasked semantic priming: Commonalities and differences. *Journal of Cognitive Neuroscience*, *25*(12), 2216–2229.

Zeelenberg, R., & Pecher, D. (2003). Evidence for long-term cross-language repetition priming in conceptual implicit memory tasks. *Journal of Memory and Language*, *49*(1), 80–94.

Zhang, J. J., & Zhang, F. L. (2014). The language representention of auditory words in Cantonese–Mandarin diglossias. *Studies of Psychology and Behavior*, *12*(4), 433–440. (In Chinese).

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Ethics declarations

Conflicts of interest

No potential conflict of interest was reported by the author.

Appendix

	Mandarin–Mandarin		Mandarin–Cantonese		Cantonese–Cantonese		Cantonese–Mandarin
1	书包-书包(bag)	1	鸡窝-鸡窦(henhouse)	1	大褛-大褛(overcoat)	1	架撑-工具(tool)
2	牙刷-牙刷(toothbrush)	2	台阶-石级(step)	2	饭羹-饭羹(meal spoon)	2	锁链-铁链(shackles)
3	电脑-电脑(computer)	3	背带-吊带(braces)	3	屋企-屋企(home)	3	花樽-花瓶(vase)
4	钢笔-钢笔(pen)	4	软饭-烂饭(soft meal)	4	滚水-滚水(boiling water)	4	银纸-钞票(bill)
5	空调-空调(air-condition)	5	手套-手袜(gloves)	5	挥春-挥春(New Year scrolls)	5	冷衫-毛衣(sweater)
6	象棋-象棋(chess)	6	电池-电芯(battery)	6	木碎-木碎(wood chips)	6	利是-红包(red envelope)
7	路灯-路灯(streetlight)	7	脸盆-面盆(washbasin)	7	蜡青-蜡青(pitch)	7	胶擦-橡皮(eraser)
8	棉花-棉花(cotton)	8	旅馆-酒店(hotel)	8	枧粉-枧粉(detergent)	8	颈链-项链(necklace)
9	彩虹-彩虹(rainbow)	9	厕所-屎坑(toilet)	9	藤条-藤条(stick)	9	衫袋-衣袋(pocket)
10	马桶-马桶(closestool)	10	晒台-天台(flat roof)	10	棉纳-棉纳(lammy)	10	光管-灯管(modulator tube)
11	书桌-书桌(desk)	11	轮子-车辘(wheel)	11	茶居-茶居(teahouse)	11	纸鹞-风筝(kite)
12	汽车-汽车(car)	12	衣服-衫裤(clothes)	12	蜜糖-蜜糖(honey)	12	啲嗒-唢呐(zurna)
13	电话-电话(telephone)	13	尺子-间尺(ruler)	13	宵夜-宵夜(snack)	13	炮仗-爆竹(firecracker)
14	鱼竿-鱼竿(fishing rod)	14	汗衫-底衫(T-shirt)	14	扣针-扣针(pin)	14	衫袖-袖子(sleeve)
15	拖鞋-拖鞋(slippers)	15	哨子-银鸡(whistle)	15	担挑-担挑(shoulder pole)	15	啤牌-扑克(poker)
16	音响-音响(acoustics)	16	扣眼-纽门(buttonhole)	16	散纸-散纸(change)	16	菲林-胶卷(film)
17	排球-排球(volleyball)	17	钱包-银包(wallet)	17	烟仔-烟仔(cigarette)	17	麻雀-麻将(mahjong)
18	喇叭-喇叭(trumpet)	18	店铺-士多(shop)	18	热痱-热痱(miliaria)	18	棉胎-棉被(quilt)
19	纸张-纸张(paper)	19	裤衩-底裤(underpants)	19	手信-手信(speciality)	19	冷饭-剩饭(leftover)
20	水杯-水杯(cup)	20	围巾-颈巾(scarf)	20	领呔-领呔(necktie)	20	汤丸-元宵(sweet dumplings)
21	枕头-枕头(pillow)	21	小刀-刀仔(knife)	21	竹排-竹排(bamboo raft)	21	腊肠-香肠(sausage)
22	筷子-筷子(chopsticks)	22	砂锅-瓦煲(marmite)	22	灯胆-灯胆(light bulb)	22	豉油-酱油(soy)
23	戒指-戒指(ring)	23	调羹-匙羹(spoon)	23	夹万-夹万(safe)	23	课室-教室(class)
24	火柴-火柴(matchstick)	24	家具-家私(furniture)	24	衣车-衣车(sewing machine)	24	梳化-沙发(sofa)
25	玻璃-玻璃(glass)	25	抽屉-柜桶(drawer)	25	铰剪-铰剪(scissor)	25	雪柜-冰箱(refrigerator)
26	手表-手表(watch)	26	扫帚-扫把(broom)	26	雪条-雪条(ice-cream)	26	雀窦-鸟窝(nest)
27	蜡烛-蜡烛(candle)	27	香皂-香枧(soap)	27	水喉-水喉(waterpipe)	27	芝士-奶酪(cheese)
28	木板-木板(board)	28	剃刀-须刨(razor)	28	恤衫-恤衫(shirt)	28	饭焦-锅巴(rice crust)
29	鼠标-鼠标(mouse)	29	钥匙-锁匙(key)	29	地毡-地毡(carpet)	29	芥辣-芥末(mustard)
30	白酒-白酒(white spirit)	30	塑料-塑胶(plastic)	30	骑楼-骑楼(arcade)	30	老泥-汗垢(sweat)
31	苹果-苹果(apple)	31	老鹰-麻鹰(eagle)	31	嫲嫲-嫲嫲(grandma)	31	契爷-干爹(nominal father)
32	袋鼠-袋鼠(kangaroo)	32	蓓蕾-花蕾(bud)	32	男仔-男仔(boy)	32	馄饨-云吞(wonton)
33	天鹅-天鹅(swan)	33	芭蕉-大蕉(plantain)	33	芽菜-芽菜(sprout)	33	苏虾-婴儿(infant)
34	树木-树木(trees)	34	柚子-碌柚(grapefruit)	34	箭猪-箭猪(porcupine)	34	烂仔-流氓(hooligan)
35	鱿鱼-鱿鱼(squid)	35	香菇-冬菇(mushroom)	35	姑丈-姑丈(uncle-in-law)	35	粟米-玉米(corn)
36	鸭子-鸭子(duck)	36	母牛-牛乸(cow)	36	姊妹-姊妹(sisters)	36	甲由-蟑螂(cockroach)
37	蚂蚁-蚂蚁(ant)	37	猴子-马骝(monkey)	37	了哥-了哥(mynah)	37	矮瓜-茄子(eggplant)
38	蜜蜂-蜜蜂(honeybee)	38	母鸡-鸡乸(hen)	38	雀仔-雀仔(bird)	38	粉葛-豆薯(yam bean)
39	蜗牛-蜗牛(snail)	39	蝙蝠-飞鼠(bat)	39	屎虫-屎虫(bug)	39	老细-老板(boss)
40	蝌蚪-蝌蚪(tadpole)	40	大腿-大髀(thigh)	40	外父-外父(father-in-law)	40	外母-岳母(mother-in-law)
41	鲜花-鲜花(flower)	41	蜘蛛-蠄蟧(spider)	41	阔佬-阔佬(rich man)	41	马蹄-荸荠(chufa)
42	斑马-斑马(zebra)	42	大蒜-蒜头(garlic)	42	家婆-家婆(mother-in-law)	42	靓仔-帅哥(handsome)
43	蝴蝶-蝴蝶(butterfly)	43	蜈蚣-百足(centipede)	43	衰人-衰人(evildoer)	43	亚婶-叔母(aunt)
44	熊猫-熊猫(panda)	44	臭虫-木虱(bug)	44	鸡仔-鸡仔(chick)	44	癫狗-疯狗(mad-dog)
45	儿童-儿童(children)	45	虱子-虱乸(louse)	45	膊头-膊头(arm)	45	盐蛇-壁虎(gecko)
46	蘑菇-蘑菇(mushroom)	46	青蛙-田鸡(frog)	46	蚊螆-蚊螆(mosquito)	46	贼佬-强盗(robber)
47	侄女-侄女(niece)	47	哑巴-哑佬(dummy)	47	薯仔-薯仔(potato)	47	亚茂-傻瓜(simpleton)
48	绵羊-绵羊(sheep)	48	义兄-契哥(sworn brother)	48	脚瓜-脚瓜(marrow)	48	乞儿-乞丐(beggar)
49	狐狸-狐狸(fox)	49	疯子-癫佬(madman)	49	番薯-番薯(sweet potato)	49	盲公-瞎子(blind)
50	燕子-燕子(swallow)	50	笨蛋-老衬(fool)	50	蒜头-蒜头(garlic)	50	家姐-姐姐(elder sister)
51	黄牛-黄牛(cattle)	51	伯父-亚伯(uncle)	51	风栗-风栗(chestnut)	51	塘尾-蜻蜓(dragonfly)
52	乌龟-乌龟(tortoise)	52	父亲-老豆(father)	52	塘虱-塘虱(catfish)	52	乌蝇-苍蝇(fly)
53	泥鳅-泥鳅(loach)	53	妻子-老婆(wife)	53	聋佬-聋佬(deaf)	53	寡佬-鳏夫(widower)
54	老虎-老虎(tiger)	54	哥哥-大佬(elder brother)	54	鼻哥-鼻哥(nose)	54	黄鳝-鳝鱼(eel)
55	辣椒-辣椒(chili)	55	水果-生果(fruit)	55	亚爷-亚爷(grandfather)	55	老母-母亲(mother)
56	喜鹊-喜鹊(magpie)	56	嫂子-亚嫂(sister-in-law)	56	脚踭-脚踭(heel)	56	狗虱-跳蚤(flea)
57	水蛇-水蛇(water snake)	57	梨子-雪梨(pear)	57	罗柚-罗柚(buttocks)	57	杨柳-柳树(willow)
58	高粱-高粱(sorghum)	58	媳妇-新抱(daughter-in-law)	58	老坑-老坑(geezer)	58	树尾-树梢(treetop)
59	表哥-表哥(cousin)	59	警察-差佬(police)	59	靓女-靓女(pretty girl)	59	细佬-弟弟(younger brother)
60	徒弟-徒弟(apprentice)	60	舅母-妗母(aunt)	60	契仔-契仔(godson)	60	白鸽-鸽子(pigeon)

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